

MISKOLC, Hungary

WMO: 127720

Lat: 48.0970N

Lon: 20.7715E

Elev: 774

StdP: 14.29

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.0	15.5	2.5	6.4	15.7	6.8	8.0	19.5	23.4	43.2	19.5	40.9	4.3	330	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	89.9	69.9	86.6	68.7	83.5	67.1	71.9	85.4	70.3	83.3	68.7	80.8	5.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.4	103.8	78.4	65.7	97.6	76.6	64.2	92.5	75.1	36.0	85.2	34.7	83.6	33.3	80.9	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.9	13.9	11.7	DB	7.6	94.2	5.0	3.4	4.0	96.6	1.1	98.6	-1.7	100.4	-5.4	102.9
			WB	6.5	74.8	4.8	2.2	3.0	76.3	0.2	77.6	-2.5	78.8	-6.0	80.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	29.9	33.3	42.4	53.1	61.3	67.6	70.9	70.4	61.4	51.4	41.4	31.4
	DBStd	16.25	7.93	7.87	7.84	7.15	6.52	6.16	5.73	5.59	6.62	7.03	7.47	7.65
	HDD50	2313	624	467	253	49	2	0	0	0	1	69	270	578
	HDD65	5573	1089	886	700	358	150	42	14	15	147	422	708	1042
	CDD50	2794	0	0	18	143	354	529	648	633	344	113	12	0
	CDD65	581	0	0	0	3	37	121	197	183	39	1	0	0
	CDH74	4913	0	0	0	34	321	1017	1696	1545	295	5	0	0
	CDH80	1577	0	0	0	2	53	305	605	545	67	0	0	0
Wind	WSAvg	5.2	4.8	5.2	6.3	5.9	5.5	5.3	5.3	4.7	4.8	4.7	4.9	4.5
Precipitation	PrecAvg	21.9	0.9	1.0	1.2	1.7	2.4	3.1	2.5	2.6	1.8	1.5	1.9	1.3
	PrecMax	33.9	2.9	3.6	3.5	3.5	5.9	7.0	5.7	6.6	7.4	7.8	4.6	3.9
	PrecMin	14.3	0.1	0.0	0.0	0.4	0.6	0.5	0.5	0.2	0.0	0.0	0.2	0.1
	PrecStd	5.1	0.8	0.9	0.8	0.9	1.1	1.6	1.2	1.7	1.4	1.5	1.3	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.3	57.2	67.4	78.1	84.9	91.1	94.9	94.5	86.2	74.1	62.4	51.1
		MCWB	44.4	46.9	52.0	60.2	65.0	71.2	70.1	69.7	67.0	61.3	54.6	45.3
	2%	DB	46.6	51.3	62.5	73.0	81.0	87.1	89.8	89.8	81.2	70.0	57.5	46.8
		MCWB	42.0	43.8	49.0	56.9	64.0	69.5	69.9	69.8	64.7	59.1	51.5	43.3
	5%	DB	43.2	47.3	58.5	69.2	77.8	83.5	86.4	85.9	77.1	66.0	54.6	43.8
		MCWB	39.7	41.8	47.4	54.7	62.6	67.3	68.7	68.7	62.9	57.4	50.0	41.1
	10%	DB	40.4	44.2	54.9	65.9	74.0	80.1	83.0	82.3	73.1	62.1	51.9	41.2
		MCWB	37.6	40.2	45.4	53.1	60.6	66.0	67.2	67.2	61.0	55.4	48.2	39.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.7	48.0	53.5	61.2	67.4	73.8	73.6	74.5	68.2	62.9	55.5	46.5
		MCDB	50.2	55.8	63.5	75.2	80.4	87.3	87.0	87.3	82.8	71.4	59.9	49.2
	2%	WB	42.8	44.7	50.5	58.6	65.3	71.0	71.7	71.6	65.9	60.4	53.0	44.0
		MCDB	45.5	49.5	59.7	70.7	78.0	83.8	85.9	84.6	78.6	68.2	56.5	46.4
	5%	WB	40.3	42.6	48.5	56.3	63.5	69.1	70.2	69.8	64.1	58.4	50.8	41.6
		MCDB	42.7	46.5	56.9	66.5	75.4	80.5	83.4	83.2	74.7	64.8	53.7	43.2
	10%	WB	37.9	40.4	46.3	54.3	61.7	67.2	68.6	68.1	62.2	56.0	48.4	39.4
		MCDB	40.2	43.8	53.4	63.6	72.3	77.7	80.2	80.5	71.2	61.2	51.5	41.1
Mean Daily Temperature Range	5% DB	MDBR	9.0	11.5	15.5	17.8	18.4	18.6	19.3	19.7	17.3	15.1	10.4	7.9
		MCDBR	11.8	16.2	21.9	23.0	23.5	23.3	24.1	24.9	23.0	20.2	14.3	11.0
		MCWBR	8.6	11.0	12.4	11.7	11.0	10.6	9.9	10.1	10.4	11.3	9.4	8.7
	5% WB	MCDBR	10.5	14.8	19.7	20.9	21.3	21.2	21.7	22.8	20.7	18.7	13.0	10.0
MCWBR		8.2	10.7	12.0	11.0	10.7	10.5	9.9	10.3	10.0	11.0	9.1	8.5	
Clear-Sky Solar Irradiance	taub		0.318	0.341	0.378	0.431	0.419	0.421	0.438	0.434	0.403	0.374	0.349	0.318
	taud		2.349	2.317	2.271	2.147	2.234	2.259	2.219	2.233	2.289	2.348	2.367	2.374
	Ebn at Noon		236	256	264	259	268	267	261	256	252	240	221	220
	Edh at Noon		23	29	36	44	43	42	43	41	35	28	23	20
All-Sky Solar Radiation	RadAvg		305	555	970	1405	1704	1833	1781	1634	1150	715	376	230
	RadStd		38	91	132	137	177	123	164	138	139	88	42	31

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (49 neighbors)	+1.08	N/A	N/A	+1.75	+1.03	+0.94	-223	-331	+176	+79

Nomenclature: See separate page